Properties of the integral of a step fun

1. additive property

$$\int_a^b (s(x)+t(x))\,dx = \int_a^b s(x)\,dx + \int_a^b t(x)\,dx, \quad \forall a,b \in R$$

proof: ① $\because$ $s+t$ step fun $\therefore$ $\int(s+t)\ \exists$.

② if $a<b$,

let P_1 be the partition of s such that s is constant on the open subintervals of P_1.

let P_2 be the partition of t ...

let $P=\{x_0, x_1, \cdots, x_n\}$ be the common refinement of P_1 and P_2.

Assume $s(x)=s_k$, $t(x)=t_k$ if $x_{k-1}<x<x_k$ $(k=1,2,\cdots,n)$

$\therefore$ $s(x)+t(x)=s_k+t_k$ if $x\in(x_{k-1},x_k)$

$\therefore$ $\int_a^b (s(x)+t(x))\,dx = \sum_{k=1}^n (s_k+t_k)(x_k-x_{k-1})$

$= \sum_{k=1}^n s_k(x_k-x_{k-1}) + \sum_{k=1}^n t_k(x_k-x_{k-1})$

$= \int_a^b s(x)\,dx + \int_a^b t(x)\,dx$

③ The case $a\geq b$ is immediate.

2. homogeneous property

$$\int_a^b c\,s(x)\,dx = c\int_a^b s(x)\,dx \quad \forall a,b,c\in R$$

proof: ① $\because$ cs step fun $\therefore$ $\int cs\ \exists$.

② if $a<b$,

let $P=\{x_0, x_1, \cdots, x_n\}$ be a partition of $[a,b]$ such that s is constant on the open subintervals of P.

Assume $s(x)=s_k$ if $x\in(x_{k-1},x_k)$ $(k=1,\cdots,n)$

$\therefore$ $cs(x)=cs_k$ if $x\in(x_{k-1},x_k)$

$\therefore$ $\int_a^b cs(x)\,dx = \sum_{k=1}^n cs_k(x_k-x_{k-1})$

$= c\sum_{k=1}^n s_k(x_k-x_{k-1})$

$= c\int_a^b s(x)\,dx$

③ The case $a\geq b$ follows directly.

3. linearity property.

$$\int_a^b (c_1 s(x)+c_2 t(x))\,dx = c_1\int_a^b s(x)\,dx + c_2\int_a^b t(x)\,dx$$

$\forall a,b,c_1,c_2\in R$

4. comparison theorem

$a<b$, $s<t \Rightarrow \int_a^b s(x)\,dx < \int_a^b t(x)\,dx$

proof: let P_1 partition of s ...

P_2 partition of t ...

P common refinement

$s(x)=s_k$, $t(x)=t_k$ if $x\in(x_{k-1},x_k)$

$\therefore$ $\int_a^b s = \sum_{k=1}^n s_k(x_k-x_{k-1})$

$< \sum_{k=1}^n t_k(x_k-x_{k-1})$

$= \int_a^b t.$

▨

5 additivity with ~~espect~~ to the interval

$$\int_a^c s(x)dx + \int_c^b s(x)dx = \int_a^b s(x)dx \quad \forall a,b,c \in R$$

proof: ① if $a<c<b$.

let P_1 partition of $[a,c]$, ...

P_2 partition of $[c,b]$, ...

$\therefore P = P_1 \cup P_2$ partition of $[a,b]$.

here $P_1 = \{x_0, x_1, \cdots, x_m\}$,

$P_2 = \{x_m, x_{m+1}, \cdots, x_n\}$

$s(x) = s_k$, if $x \in (x_{k-1}, x_k)$

$(k = 1, 2, \cdots, m, \cdots, n)$

$P = P_1 \cup P_2 = \{x_0, x_1, \cdots, x_m, x_{m+1}, \cdots, x_n\}$

$$\therefore \int_a^c s + \int_c^b s = \sum_{k=1}^{m} s_k (x_k - x_{k-1}) + \sum_{k=m+1}^{n} s_k (x_k - x_{k-1})$$

$$= \sum_{k=1}^{n} s_k (x_k - x_{k-1})$$

$$= \int_a^b s(x)dx$$

② $\forall a,b,c \in R$, see 02/13.

6. invariance under translation

$$\int_a^b s(x)dx = \int_{a+c}^{b+c} s(x-c)dx, \quad \forall a,b,c \in R$$

proof: ① if $a<b$

let $P = \{x_0, x_1, \cdots, x_n\}$ partition of $[a,b]$,

$\therefore P' = \{x_0+c, x_1+c, \cdots, x_n+c\}$ partition of $[a+c, b+c]$.

let $s(x) = s_k$ if $x \in (x_{k-1}, x_k)$

$\therefore s_1(x) := s(x-c) = s_k$, if $x \in (x_{k-1}+c, x_k+c)$

$$\therefore \int_a^b s(x)dx = \sum_{k=1}^{n} s_k (x_k - x_{k-1})$$

$$= \sum_{k=1}^{n} s_k ((x_k+c) - (x_{k-1}+c))$$

$$= \int_{a+c}^{b+c} s_1$$

$$= \int_{a+c}^{b+c} s(x-c)dx$$

② if $a \geq b$, the proof is easy.

7. expansion or contraction of the interval

$$\int_{ka}^{kb} s\left(\frac{x}{k}\right)dx = k\int_a^b s(x)dx, \quad \forall a,b \in R, k \neq 0$$

(see 02/13)